

Legal Whitepaper

On-Chain Liquidity and Price Discovery for Tokenized Life Sciences Milestone Payment Rights

Aditya Bhosale
University of Pennsylvania

April 2026

DISCLAIMER

This document is a research analysis prepared by the author, an undergraduate student at the University of Pennsylvania, in connection with independent academic research on mechanism design, tokenized real-world assets, and prediction market infrastructure. It does not constitute legal advice, an offer to sell or solicitation of an offer to buy any security, or a recommendation to engage in any transaction. The analysis reflects the author's interpretation of publicly available statutes, regulations, and precedent as of the date of writing; it has not been reviewed or endorsed by licensed legal counsel, and the author is not an attorney. Regulatory treatment of the instruments and structures discussed herein is actively evolving and may differ materially from the analysis presented. Any party contemplating the design, issuance, purchase, sale, or trading of instruments along the lines described in this document should obtain independent legal advice from a licensed securities attorney and should not rely on any statement made herein.

Abstract

This memo outlines the legal and architectural framework for an on-chain liquidity and price discovery layer for tokenized life sciences milestone payment rights. The underlying asset — a contractual right to receive payment upon a specific clinical event (FDA approval, Phase II trial success) — is structured into a Delaware special purpose vehicle (SPV) and tokenized as a security. A Liquidity-Sensitive Logarithmic Market Scoring Rule (LS-LMSR) automated market maker (AMM) runs as a separate prediction market layer on top, providing continuous price discovery between clinical catalysts. This memo addresses six open legal and architectural questions: asset classification, exemption structure, SPV architecture and permissioning, settlement stack, the two-layer legal separation problem, and material non-public information (MNPI) risk.

1 Asset Classification

The threshold legal question is whether a tokenized milestone payment right constitutes a security under the *Securities Act of 1933*. Under the *Howey* test, an instrument is a security if it involves (1) an investment of money, (2) in a common enterprise, (3) with an expectation of profit, (4) derived from the efforts of others.

A tokenized milestone payment right satisfies all four prongs. Investors commit capital to the SPV (prong 1); the SPV pools that capital against a single underlying contractual claim (prong 2);

investors expect to receive a cash payout if the clinical milestone is achieved (prong 3); and that payout depends entirely on the efforts of the pharma or biotech counterparty conducting the trial (prong 4). The instrument is therefore a security under current doctrine.

The more contested question is whether the prediction market layer — outcome shares (YES/NO tokens) that reference the underlying milestone — also constitutes a security, or whether it falls under Commodity Futures Trading Commission (CFTC) jurisdiction as an event contract. Binary options on single events have been treated as securities by the SEC in prior enforcement actions (*SEC v. Banc de Binary*, 2013), but event contracts on non-financial underlying events have been regulated by the CFTC in others (*In re Intrade*, 2012). No controlling precedent exists for a two-layer instrument where the prediction market layer references a tokenized security as its underlying. This remains the most significant unresolved legal question in the model’s architecture.

2 Exemption Structure

Given that the SPV token is a security, the offering must either be registered with the SEC or qualify for an exemption. Three exemptions are relevant:

Regulation D, Rule 506(c) permits general solicitation of accredited investors without SEC registration. Rule 506(c) is the most practical near-term exemption because it allows broad outreach while restricting participation to verified accredited investors. The primary constraint is that SPV interests issued under Rule 506(c) are “restricted securities” subject to a six-month holding period under Rule 144 before resale. This creates direct tension with the prediction market layer, which requires continuous trading of outcome shares — an issue addressed in Section 5.

Regulation S exempts offerings made exclusively to non-US persons from SEC registration. Reg S is operationally viable — Robinhood launched tokenized equities for European users under a similar structure — but limits the market to non-US participants, constraining liquidity and excluding the most natural institutional counterparties (US pharma companies and US-based royalty funds).

Regulation A+ permits offerings of up to \$75 million to both accredited and non-accredited investors following SEC qualification. Reg A+ is the most democratizing structure but requires full SEC qualification upfront, making it the least practical for an early-stage deployment.

The recommended near-term structure is Reg D 506(c), with Reg S as a parallel offshore track for non-US participants. This mirrors the structure used by Securitize for institutional tokenized fund offerings and is the most immediately deployable without requiring SEC registration.

3 SPV Architecture and Permissioning

3.1 SPV Structure

The underlying milestone payment right is held by a Delaware limited liability company (LLC) established as a special purpose vehicle. The SPV has a single purpose: to hold the contractual right to receive payment from the pharma or biotech counterparty upon achievement of the specified clinical milestone. Token holders own membership interests in the SPV, which in turn represent a proportional economic claim on the milestone payment.

The SPV structure establishes the on-chain token as the authoritative legal record of ownership, rather than a digital representation of an off-chain claim. This requires explicit drafting in the SPV operating agreement to ensure that the token register — maintained on-chain via the ERC-3643 compliance module — constitutes the definitive record of membership interests, superseding any off-chain cap table.

3.2 Token Standard: ERC-3643

The SPV membership interest token is implemented using the ERC-3643 (T-REX) standard, which builds compliance logic directly into the token contract. ERC-3643 enforces transfer restrictions on-chain through four components:

- **Identity Registry:** maps wallet addresses to verified identity claims (accredited investor status, jurisdiction, sanctions screening)
- **Compliance Module:** smart contract that checks the identity registry on every transfer and enforces Reg D lock-up periods and jurisdiction restrictions
- **Token Contract:** the ERC-3643 asset token, with transfer hooks calling the compliance module
- **Claims Issuer:** the trusted entity (SPV administrator or third-party KYC provider) that signs identity attestations

This architecture satisfies Reg D’s transfer restriction requirements without requiring off-chain compliance middleware for every transaction. It is the same standard used by Securitize for BlackRock’s BUIDL fund and KKR’s tokenized private equity offerings.

3.3 Modular Permissioning

The permissioning architecture is designed to be modular — separating the venue-level access control (who can participate in the prediction market) from the asset-level transfer restrictions (who can hold the SPV token). This separation is architecturally necessary because the two layers may operate under different regulatory regimes (see Section 5) and have different liquidity requirements.

Venue-level permissioning is enforced at the prediction market smart contract layer on Base: only wallets that have passed KYC and hold a valid identity attestation can interact with the AMM. Asset-level permissioning is enforced by the ERC-3643 compliance module on Ethereum mainnet: only wallets with verified accredited investor status can hold or receive the SPV token.

4 Settlement Stack

4.1 Chain Architecture

The model operates across two chains with distinct roles:

- **Ethereum Mainnet:** SPV token issuance, ERC-3643 compliance enforcement, legal ownership registry
- **Base (Coinbase L2):** Prediction market AMM, continuous outcome share trading, USDC settlement of resolved markets

This separation reflects both the technical and legal architecture. Ethereum mainnet provides the institutional trust, security, and ERC-3643 ecosystem maturity required for a regulated security. Base provides the low transaction costs and USDC-native infrastructure required for continuous prediction market trading.

4.2 Settlement Bridging

When the prediction market resolves on Base, USDC distributions to outcome share holders execute natively on Base. Settlement of the underlying SPV milestone payment — the transfer of the actual contractual cash flow from the pharma counterparty to SPV token holders — requires a bridge between the oracle confirmation on Base and the ERC-3643 token registry on Ethereum mainnet. Circle’s Cross-Chain Transfer Protocol (CCTP) provides the USDC transfer mechanism; the legal finality of the settlement event is recorded on Ethereum mainnet via a state update to the SPV’s token contract.

4.3 Legal Recognition of On-Chain Settlement

On-chain settlement finality is technically irreversible but not automatically legally recognized. For institutional participants, the settlement event must be recognized by the SPV operating agreement as constituting final discharge of the milestone payment obligation. This requires explicit drafting: the SPV operating agreement must designate the smart contract execution event as the legally binding settlement mechanism, subject to the oracle resolution outcome specified in Section 5.

5 Two-Layer Separation and the Composability Question

5.1 Legal Separation of the Prediction Market Layer

The central structural question is whether the prediction market layer (outcome shares) can be legally separated from the underlying SPV token (the security) such that outcome shares fall outside SEC jurisdiction. If outcome shares are deemed a disguised resale of the restricted SPV interest, they would be subject to the same Reg D transfer restrictions — making continuous trading legally impermissible during the lock-up period.

The legal basis for separation rests on two arguments. First, outcome shares do not convey ownership of the SPV or any claim on its assets — they are binary instruments that pay out based on a public event, not on the SPV’s economic performance. Second, the prediction market is structured as a CFTC-regulated event contract rather than a securities offering: participants are not investing in a common enterprise but rather taking positions on a binary outcome. Whether regulators would accept this characterization or look through the structure to find an indirect securities offering remains the most significant unresolved legal question.

5.2 Soulbound vs. Composability Tradeoff

The composability of outcome shares — whether they can be freely transferred, used as collateral, or integrated into DeFi protocols — presents a fundamental design tradeoff:

Soulbound tokens (non-transferable) solve the Reg D resale problem cleanly. If outcome shares cannot be transferred, they cannot constitute a secondary market in the restricted SPV interest. The legal exposure is substantially reduced. The cost is the elimination of secondary market liquidity and DeFi composability — outcome shares cannot be used as collateral, traded on secondary venues, or integrated into lending protocols.

Composable tokens (freely transferable) enable DeFi integration: outcome shares can trade on decentralized exchanges, serve as collateral in lending protocols, and be integrated into structured products. This maximizes liquidity and capital efficiency. The legal risk is that freely transferable outcome shares referencing a restricted security may be deemed an unlawful secondary market, triggering SEC enforcement.

The resolution of this tradeoff depends on the legal characterization of outcome shares in Section 5.1. If outcome shares are successfully characterized as CFTC event contracts, composability is permissible. If they are deemed securities, soulbound or transfer-restricted implementation is required. The model’s architecture is designed to support either implementation, with the composability decision deferred pending legal opinion.

6 MNPI Risk and Oracle Data Policy

6.1 The MNPI Problem

The LS-LMSR prior is seeded from a computational model incorporating proprietary preclinical data provided by the pharma counterparty. This data — unpublished assay results, internal efficacy models, toxicity findings — almost certainly constitutes material non-public information (MNPI) if it relates to a publicly traded company’s drug pipeline. Any market participant who trades outcome shares with knowledge of how the prior was constructed may be trading on MNPI, creating insider trading liability under Rule 10b-5.

6.2 Primary Design Response: Private Company Counterparties

The near-term design response is to restrict the model to milestone payment rights from private biotechs — companies not subject to public company securities laws. Private company MNPI does not implicate Rule 10b-5, which applies to trading in the securities of publicly traded issuers. A private biotech’s unpublished preclinical data, while commercially sensitive, does not create the same insider trading exposure as data relating to a public company’s pipeline.

This constraint is compatible with the model’s primary use case. Development-stage private biotechs seeking non-dilutive financing through milestone payment tokenization are the most natural initial counterparty profile. Publicly traded pharma companies, which have more complex MNPI obligations, are excluded from the near-term deployment.

6.3 Long-Term Design Response: Information Barrier Architecture

As the model scales to include public company counterparties, an information barrier architecture analogous to investment bank research division protocols would be required. Under this structure, proprietary pharma data feeds into the prior calculation through a trusted intermediary operating under strict confidentiality obligations, and market participants see only the resulting price output — not the underlying data inputs. This prevents the transmission of MNPI from the data provider to market participants while preserving the quality of the computational prior.

The design of a legally defensible information barrier for an on-chain oracle is an open architectural question. Existing solutions (UMA’s optimistic oracle, Pyth’s multi-source price aggregation) are designed for continuous price feeds and are not directly applicable to bespoke clinical event data. A purpose-built oracle architecture with formal information barrier protocols is a prerequisite for public company counterparty onboarding.

7 Open Questions

The following questions remain unresolved and are the subject of ongoing legal and technical development:

1. **SEC vs. CFTC jurisdiction over outcome shares:** Whether binary outcome shares referencing a tokenized security fall under SEC or CFTC jurisdiction, and whether the two-layer structure is legally respected or collapsed by regulators.
2. **Oracle resolution and legal finality:** Which oracle architecture provides legally recognized settlement trigger authority, and how dispute resolution integrates with the SPV's contractual payment obligation.
3. **Soulbound vs. composability:** The final composability decision is contingent on the jurisdictional determination in question 1.
4. **SPV assignment without counterparty consent:** Whether the pharma counterparty's assignment of a milestone payment right to the SPV requires regulatory consent (FDA, IRB) given that the underlying right is tied to a regulated clinical process.
5. **Rule 144 resale and the prediction market layer:** Whether continuous trading of outcome shares during the Reg D lock-up period constitutes an unlawful secondary market in the restricted SPV interest.